

dieT : A Diet Planning Expert System

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Aim : To represent knowledge using Prolog by implementing a small expert system for personalized diet planning based on health conditions, dietary preferences, and user profiles.

1. Problem Statement

Many individuals struggle to find appropriate dietary recommendations tailored to their specific health conditions and preferences. General dietary advice often fails to account for individual differences, medical conditions, and dietary restrictions. This project addresses the need for personalized nutrition guidance by leveraging knowledge representation techniques to create an intelligent diet planning system.

2. Introduction

HealthWiseDiet is a knowledge-based expert system that provides personalized diet recommendations based on a user's health profile. The system combines rule-based reasoning through Prolog with web technologies to deliver an accessible and user-friendly diet planning application. It addresses the challenge of dietary decision-making for individuals with specific health conditions such as diabetes, hypertension, heart disease, or obesity, while also accommodating dietary preferences such as vegetarian or vegan diets.

3. Modules/Features

1. User Authentication & Profile Management
 - Registration & Login: Users can create an account with personal details (name, email, age, gender, weight, height, etc.) and securely log in.
 - Profile Editing: Users can update their health information and preferences at any time, ensuring recommendations remain relevant.
 - Session Management: The app uses Flask-Login to manage user sessions, ensuring secure access to personalized features.
2. Rule - Based Diet Recommendation Engine (Expert System)
 - Prolog Knowledge Base (kb.pl): The core dietary logic is encoded in Prolog, representing nutrition rules, food groups, and health condition guidelines.
 - Python-Prolog Integration: The diet_planner.py module uses the pyswip library to query the Prolog knowledge base from Python, passing user data and retrieving recommended and avoided foods.
 - Personalization: Recommendations are tailored based on the user's age, gender, BMI, health conditions (like diabetes, hypertension), and dietary preferences (vegetarian, vegan, etc.).

3. AI-Enhanced Advice (Google Gemini Integration)
 - Gemini AI Module (gemini_ai.py): After generating initial recommendations, the app sends user data and Prolog results to Google Gemini AI.
 - Advanced Insights by Gemini AI: It provides a brief health profile analysis, key nutritional insights, lifestyle tips, meal timing advice, supplement suggestions
4. Personalized Meal Plan Generation
 - Meal Planning Logic: The app organizes recommended foods into daily meal plans (breakfast, lunch, dinner, snacks) using food group information from the knowledge base.
 - Randomization & Variety: Meal plans are varied and balanced, ensuring users get a diverse diet while adhering to their restrictions and preferences.
5. Plan Management
 - Save Plans: Users can save their personalized diet plans for future reference.
 - View & Delete Plans: Saved plans can be reviewed or deleted as needed, allowing users to track their dietary journey over time.
6. UI and Experience
 - Flask Templates: The app uses HTML templates for all pages (home, login, registration, profile, recommendations, meal plans, etc.), ensuring a smooth and intuitive user experience.
 - Feedback & Notifications: Users receive feedback (success, error, info messages) for actions like login, registration, saving plans, etc.
7. Database Management
 - SQLAlchemy ORM: All user data, preferences, and saved plans are stored securely in a database using SQLAlchemy, ensuring data integrity and easy retrieval.

4. Dataset Information

The system utilizes Prolog knowledge base (kb.pl) as the core dataset. This dataset was derived from rendering the following kaggle datasets.

- [Nutritional Facts of Common Foods](#)
- [Nutritional Food Facts](#)

The above 2 databases give us the food along with its type. Eg milk, yogurt, cheese are dairy products and cabbage, beans and spinach are vegetables. This data is used to make facts in prolog. Eg *food_group(milk, dairy)*. *food_group(palak, vegetables)*.

- [Food Suitable for Diabetes](#)

This dataset is used to determine and list down the food suitable for diabetes in the form of facts in prolog. Eg. Brown rice and flax seeds are some items recommended to people with diabetes and the facts in prolog as written as follows: *condition_food(diabetes, brown_rice, recommended)*. *condition_food(diabetes, flaxseeds, recommended)*.

- [Daily Meal Nutrition and Diseases](#)

This dataset contains breakfast, lunch, snacks and dinner recommendations for people based on the diseases they have. Eg. A person with hypertension and heart disease can eat paneer with veggies for breakfast, lentil soup for lunch, apple for snacks and vegetable stir-fry with brown rice

for dinner. These diet recommendations along with the diseases are backtracked to make the database.

- 200+ food items categorized by health impact and nutritional properties
- Dietary rules for health conditions (diabetes, hypertension, heart disease, obesity)
- Food classification for dietary preferences (vegetarian, vegan, non-vegetarian)
- Age-based nutritional recommendations
- BMI-based dietary guidelines
- Food groupings for meal structuring (proteins, grains, vegetables, fruits, etc.)

The knowledge base is particularly focused on Indian cuisine, making it culturally relevant and practical for users familiar with these food items.

5. Code

The application is built using:

- Backend: Flask (Python web framework)
- Knowledge Representation: SWI-Prolog with PySwip for Python integration
- Database: SQLAlchemy ORM with SQLite
- Authentication: Flask-Login
- Form Handling: Flask-WTF
- AI Enhancement: Google's Gemini AI API
- Frontend: HTML, CSS, JavaScript with Bootstrap

Key files include:

- `app.py`: Main Flask application with routes and controllers
- `kb.pl`: Prolog knowledge base defining food-condition relationships and rules
- `diet_planner.py`: Interface between Python and Prolog for recommendations
- `gemini_ai.py`: Integration with Google's Gemini AI for enhanced insights
- `models.py`: Database models for user profiles and saved plans
- `forms.py`: Form definitions for data collection and validation

6. Results

The system generates personalized diet recommendations based on factors:

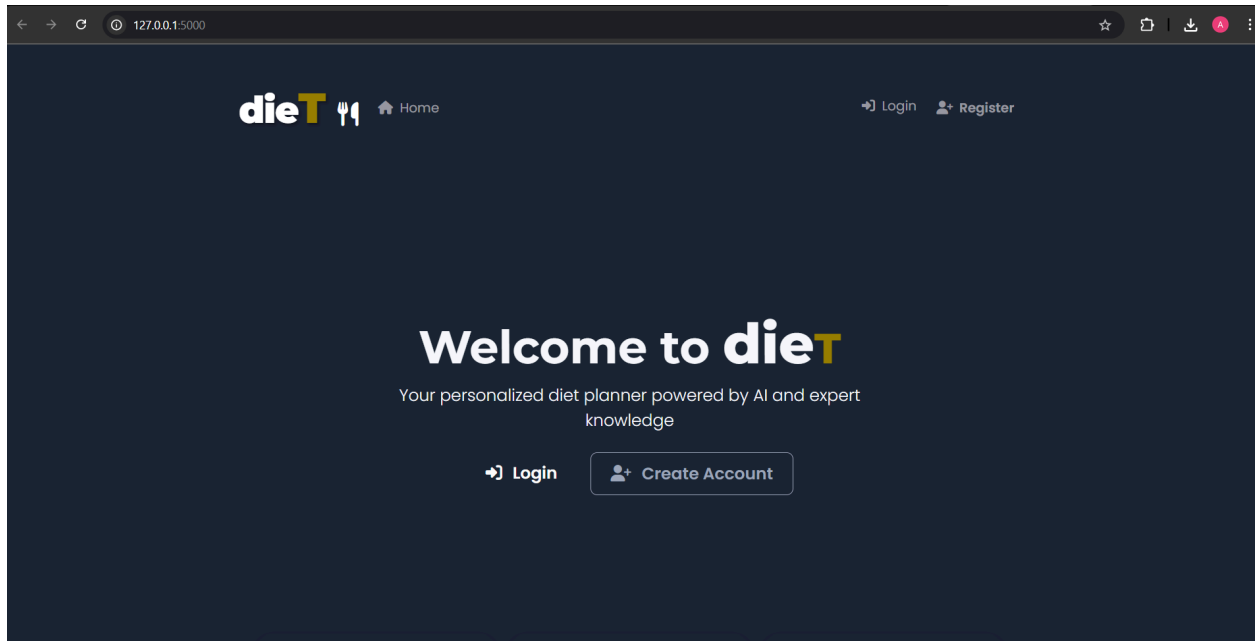
- Health conditions and their dietary implications
- Dietary preferences and restrictions
- Age and gender-specific nutritional needs
- BMI and weight management considerations
- Activity levels and lifestyle factors

The recommendations are presented as:

1. Lists of recommended and avoided foods
2. Structured meal plans divided into breakfast, lunch, dinner, and snacks
3. AI-enhanced nutritional insights and lifestyle tips
4. Personalized advice on meal timing and supplements

7. Screenshots

- Home/landing page



- User registration page

A screenshot of a web browser showing the 'Create Account' registration form. The browser's address bar displays '127.0.0.1:5000/register'. The form is titled 'Create Account' with a user icon. It contains several input fields: 'Username' (filled with 'aditihinge'), 'Email' (filled with 'aditi.hinge@cumminscollegi'), 'Password' (masked with dots), and 'Confirm Password' (masked with dots). Below these is a section titled 'Profile Information (Optional)' with fields for 'First Name' (filled with 'Aditi'), 'Last Name' (filled with 'Hinge'), 'Age' (filled with '22'), 'Gender' (filled with 'Female'), 'Weight (kg)' (filled with '54'), and 'Height (cm)' (filled with '157'). A 'Register' button is at the bottom right of the form.

- Login Interface

A screenshot of a web browser showing the 'Login' interface. The browser's address bar displays '127.0.0.1:5000/login'. The page features a header with the 'dieT' logo, a home icon, and links for 'Login' and 'Register'. The main content area contains a 'Login' form with a title bar. The form has fields for 'Username' (filled with 'aditihinge') and 'Password' (masked with dots). Below the password field is a 'Remember Me' checkbox. A 'Sign In' button is positioned below the checkbox. At the bottom of the form, there is a link: 'Don't have an account? [Register](#)'.

- Health profile input form

127.0.0.1:5000/plan

Daily Water Intake *

4-8 cups

Food Allergies/Intolerances *

☒ None ☐ Dairy ☐ Gluten

☐ Nuts ☐ Shellfish ☐ Soy

Meal Timing Preference *

Regular (3 meals per day)

Health Goals *

☐ None ☐ Weight Loss ☒ Muscle Gain

☐ Better Sleep ☐ More Energy ☒ Manage Health Condition

Current Supplements *

☐ None ☐ Multivitamin ☒ Vitamin D

☐ Omega-3 ☐ Protein ☐ Probiotics

Generate Diet Plan

- Results page with recommendations

127.0.0.1:5000/recommendations

Your Health Profile

Age: 22 years **BMI:** 21.9 (Normal)

Weight: 54.0 kg **Health Conditions:**

Height: 157.0 cm • Diabetes

Gender: Female • Heart disease

Dietary Preferences:

• Non_vegetarian

• Vegetarian

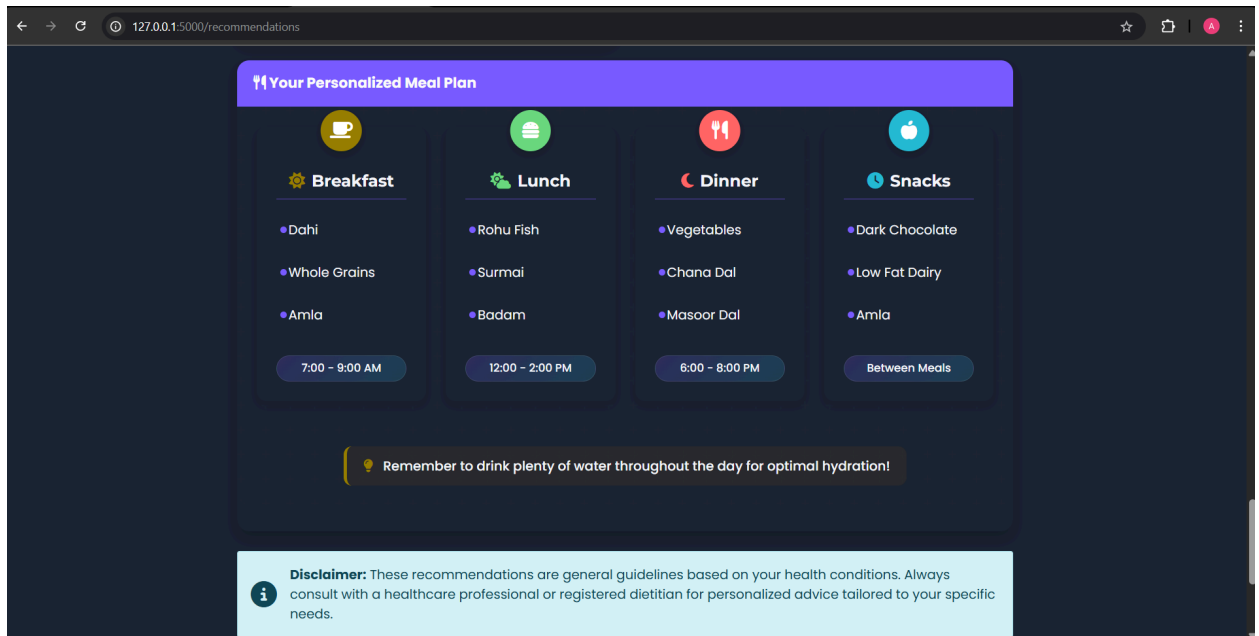
Recommended Foods

- ✓ Akhrot
- ✓ Amla
- ✓ Badam
- ✓ Bajra
- ✓ Balanced Diet

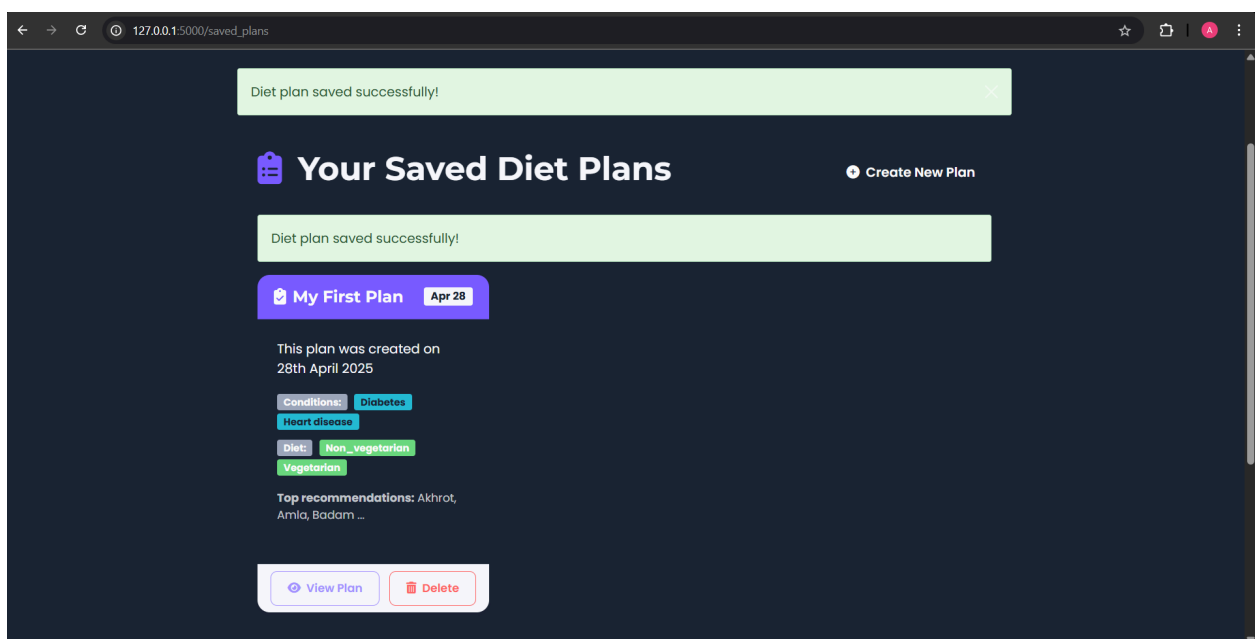
Foods to Avoid

- ✗ Aamras
- ✗ Alcohol
- ✗ Bhajias
- ✗ Excessive Alcohol
- ✗ Excessive Caffeine

- Meal plan visualization



- Saved plans dashboard



8. Conclusion

The HealthWiseDiet expert system demonstrates the effective application of knowledge representation techniques using Prolog for solving real-world health management challenges. By combining symbolic AI through rule-based reasoning with modern web technologies and natural language processing, the system delivers personalized dietary guidance that accounts for complex interrelationships between health conditions, food properties, and user preferences. Future enhancements could include expanding

the knowledge base to cover more international cuisines, incorporating more sophisticated reasoning about nutrient interactions, and developing a mobile application for improved accessibility. The project shows promising potential as a practical tool for individuals seeking tailored dietary guidance based on their specific health needs.